# Structural Closure of Global Resolution Geometry

## Recognized Frameworks that Replace Sampling with Derivation: o-Minimality, Nerves and Descent, Normal Forms, and Cone Contraction

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**Abstract**

Čencov's theorem convinces because it is a derivation inside a recognized framework, not because of examples. Many global statements of RG have so far been supported by finite synthetic batteries: the wall-crossing gauges, the atlas, the mesh closure, the holonomy counts and the forgetting rates. We show that four recognized mathematical frameworks let most of them be stated and proved structurally. Their standard theorems are imported, and the RG corollaries are proved here.

(i) *o-Minimality (tame geometry).* All RG objects of a definable experiment are definable: fibres, walls, the Fisher field and the identifiability modulus. Consequently:

- there are finitely many fibre types, with a finite atlas (Hardt, cell decomposition);
- a gauge crosses a singular point continuously iff it has one limit along every *definable curve* (curve selection). This turns the print's ray-wise canary into a theorem: rays fail, curves suffice;
- identifiability moduli are eventually monotone, and are power laws with rational exponents in polynomially bounded structures (Miller, Łojasiewicz).

Tameness is necessary: a $\sin(1/\theta)$ family has infinitely many walls.

(ii) *Nerves and descent.* For a good cover, the finite nerve computes every global invariant exactly. RG's coverage thresholds are goodness conditions, and the obstruction tower is standard obstruction theory.

(iii) *Normal forms.* Real symmetric degeneracies have codimension 2. Crossings are avoided in one-parameter families, and the conical normal form forces sign monodromy $-1$. Protocol invariants are constant along strata (Thom).

(iv) *Cone contraction (Birkhoff–Hopf).* Forgetting is a strict contraction of Hilbert's projective metric with the closed-form coefficient $\tanh(\Delta/4)$, e.g. $1 - 2\varepsilon$ for a binary symmetric channel. This holds uniformly for *forced*, time-inhomogeneous protocols of bounded projective diameter. For piecewise expanding chaos, Lasota–Yorke gives a spectral gap.

**Evidence and scope.** Six exact or symbolic witnesses, with no random sampling, check transcription and constants. A ledger maps each synthetic battery to its structural replacement. The paper makes no novelty claim for the frameworks; its contribution is placing RG inside them, which makes RG's global layer easier to follow and harder to dispute.

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# 1 The standard: derivation inside a recognized framework

A closure is *structural* if it consists of three things:

(i) a theorem of a recognized framework, cited;

(ii) a proof that RG's objects satisfy that framework's hypotheses;

(iii) a proof of the RG corollary.

Finite computations may only *witness* constants and transcriptions. They never carry the claim.

| RG global issue | Framework | Structural statement |
|---|---|---|
| Finiteness of types, the atlas, tame issues (L132) | o-minimality | Hardt triviality and cell decomposition: finitely many types, a finite atlas |
| Wall crossing at singular points (L129–L131, ray-wise canary) | o-minimality | curve selection: crossing iff definable-curve limits agree |
| Rates and moduli at walls | o-minimality | monotonicity, Miller dichotomy, Łojasiewicz: power laws |
| Mesh closure, atlas gluing (A6, mesh paper) | nerves, Čech descent | nerve theorem; cocycle classes; Steenrod obstruction theory |
| Invariants change only at walls | stratification | Thom's first isotopy lemma |
| Forced monodromy (L122) | normal forms | von Neumann–Wigner codimension 2; conical form; monodromy $-1$ |
| Forgetting, forced protocols, chaos (L53–L55, L133) | cone geometry, transfer operators | Birkhoff–Hopf $\tanh(\Delta/4)$; Doeblin; Lasota–Yorke gap |

# 2 Tame geometry: o-minimality

**Definition 2.1** (Tame experiment)**.** An experiment $\theta \mapsto p_\theta \in \Delta_n$, $\theta \in \Theta \subset \mathbb{R}^d$, is *tame* if $\Theta$ and the law map are definable in an o-minimal expansion of the real field. Examples are:

- semialgebraic families;
- restricted analytic families ($\mathbb{R}_{\mathrm{an}}$);

- exponential families, which are definable in $\mathbb{R}_{\exp}$ (Wilkie [3]).

**Lemma 2.2** (RG objects are definable)**.** *For a tame experiment the following are definable:*

- *the observational equivalence relation $\theta \sim \theta' \iff p_\theta = p_{\theta'}$;*
- *its fibres;*
- *the Fisher field, where the law map is $C^1$;*
- *the wall set where the rank of $Dp$ drops;*
- *for any definable observation metric, the identifiability modulus $\omega(\varepsilon) = \sup\{\|\theta - \theta'\| : d_{\mathrm{obs}}(p_\theta, p_{\theta'}) \le \varepsilon\}$.*

*Proof.* Definable sets are closed under Boolean operations, projections and products. Derivatives of definable $C^1$ functions are definable. The rank-drop set is cut out by minors. A supremum over a definable family is definable, because $\{(\varepsilon, s) : \exists \theta, \theta' \, (\cdots \wedge \|\theta - \theta'\| > s)\}$ is definable. □

**Imported Theorem 2.3** (Hardt triviality; cell decomposition [1, 2])**.** *Let $f : X \to Y$ be a definable map. Then $Y$ has a finite definable partition over each piece of which $f$ is definably trivial. Every definable set has a finite decomposition into cells.*

**Theorem 2.4** (Finite types and finite atlas)**.** *For a tame experiment:*

*(a) the observational fibres fall into finitely many types up to definable homeomorphism;*

*(b) $\Theta$ has a finite definable atlas whose charts are cells on which the rank of $Dp$ is constant;*

*(c) the walls, i.e. the union of the frontiers of the rank strata of $Dp$, form a definable set of dimension less than $d$.*

*Proof.* (a) Apply Hardt to the projection of the graph of $\sim$ onto $\Theta$, using Lemma 2.2. (b) Apply cell decomposition compatible with the definable rank strata. (c) The rank strata $S_r = \{\theta : \operatorname{rank} Dp_\theta = r\}$ are finitely many definable sets. By the frontier inequality $\dim(\overline{S_r} \setminus S_r) < \dim S_r \le d$ [1], each frontier, and so their finite union, has dimension less than $d$. □

*No-Go* 2.5 (Tameness is necessary; W6). The non-definable family $p_\theta = (\frac{1}{2} + \frac{1}{4}\sin(1/\theta), \frac{1}{2} - \frac{1}{4}\sin(1/\theta))$ has walls at $\theta = 1/(\pi k)$ for every $k$, accumulating at 0. No finite atlas exists near 0. By contrast, the semialgebraic family of root counts of $t^2 + at + b$ has exactly three types, separated by the wall $a^2 = 4b$ (W6).

*Boundary.* Theorem 2.4 gives finiteness *per family*. A uniform enumeration over all families of a given complexity, which is L132 and of Hilbert-16 type, is not implied.

**Imported Theorem 2.6** (Curve selection [1])**.** *If $A$ is definable and $x_0 \in \overline{A} \setminus A$, then there is a definable continuous curve $\gamma : (0, 1) \to A$ with $\gamma(t) \to x_0$.*

**Theorem 2.7** (Crossing criterion)**.** *Let $b$ be definable and bounded on a punctured neighbourhood of $x_0$. Then $b$ extends continuously to $x_0$ iff $\lim_{t \to 0} b(\gamma(t))$ exists and is the same for every definable curve $\gamma \to x_0$. Limits along rays do not suffice.*

*Proof.* ($\Rightarrow$) This is immediate.

($\Leftarrow$) Let $M(r) = \sup_{|x - x_0| = r} b$ and $m(r) = \inf_{|x - x_0| = r} b$. Both are definable functions of one variable, hence eventually monotone, with limits $M_0 \ge m_0$. If $M_0 = m_0$, the uniform limit exists. If $M_0 > m_0$, put $c = (M_0 + m_0)/2$. The definable sets $\{b > c + \delta\}$ and $\{b < c - \delta\}$, for small $\delta$, both have $x_0$ in their closure. Curve selection gives two definable curves whose limits differ, a contradiction.

*Rays fail* (W3): for $b = xy^2/(x^2 + y^4)$ every ray, including the $y$-axis, gives 0, while the definable curve $x = y^2$ gives $\frac{1}{2}$. □

This is the structural form of the print's uniform-limit criterion and its ray-wise no-go. Checking every definable curve can still be infinite in general, but in semialgebraic families the curves can be taken to be Puiseux arcs, which form a finite-parameter search.

**Imported Theorem 2.8** (Monotonicity; Miller's dichotomy; Łojasiewicz [1, 4, 5])**.** *A definable function of one variable is eventually monotone and continuous. In a polynomially bounded o-minimal structure, every definable $\omega : (0, \epsilon_0) \to \mathbb{R}$ satisfies $\omega(\varepsilon) \sim c\,\varepsilon^r$, with $r$ in the field of exponents, which is $\mathbb{Q}$ for semialgebraic sets and $\mathbb{R}_{\text{an}}$. Definable $f \geq 0$ with zero set $Z$ satisfies $f(x) \geq c\,\text{dist}(x, Z)^\alpha$ locally.*

**Theorem 2.9** (Moduli are power laws)**.** *For a tame experiment in a polynomially bounded structure, the identifiability modulus satisfies $\omega(\varepsilon) \sim c\,\varepsilon^r$ as $\varepsilon \to 0$ with $r \in \mathbb{Q}$, or $\omega$ is bounded away from 0 (a persistent ambiguity). At a wall, a Łojasiewicz exponent $\alpha$ of $d_{\text{obs}}$ gives $\omega(\varepsilon) \leq C\varepsilon^{1/\alpha}$.*

*Proof.* $\omega$ is definable (Lemma 2.2), so Miller's dichotomy applies. Apply Łojasiewicz to $f(\theta) = d_{\text{obs}}(p_\theta, p_{\theta_0})$, whose zero set is the fibre. □

*Witness (W5).*

- At the singular Bernoulli wall, TV $= \theta^2$, so $\omega = \varepsilon^{1/2}$.
- For the fixed-spectrum eigenline (spectral paper), $\omega = \varepsilon/g$, i.e. $r = 1$.

Together with the Rooting Closure's RLCT invariance, wall rates are now fully structural for tame models.

## 3 Global gluing: nerves, descent and obstruction theory

**Imported Theorem 3.1** (Nerve theorem [6, 7])**.** *If an open cover of a paracompact space is good, i.e. every nonempty finite intersection is contractible, then the union is homotopy equivalent to the nerve of the cover.*

**Corollary 3.2** (Exact mesh and atlas closure)**.** *If RG's charts, or the cells of a mesh, form a good cover, every homotopy invariant of the union can be computed exactly from the finite nerve. This includes $H^1$, $\pi_1$ classes, and the holonomy classes of flat bundles with discrete structure group. RG's coverage conditions are sufficient conditions for goodness. For example, consecutive arcs on a circle of line directions whose overlaps are single intervals give a good cover. Therefore the mesh closure of the mesh paper and the exact holonomy thresholds of the spectral paper are instances of the nerve theorem.*

*Proof.* Apply the nerve theorem. Flat bundles with discrete structure group are classified by $\text{Hom}(\pi_1, G)/G$, which is a homotopy invariant. On a circle, an open arc of length below the period is contractible, and the intersection of two consecutive arcs under the coverage condition is a single open arc. □

*Witness (W2).* Covers of the circle by $N \geq 3$ arcs, with only consecutive overlaps, give a nerve with $H^1 = \mathbb{Q}$, as for the circle. Two arcs overlap in two components, so the cover is not good, and the nerve, a single edge, gives $H^1 = 0$. This is the structural form of the aliasing canary.

**Imported Theorem 3.3** (Čech descent; obstruction theory [8, 9])**.** *Local sections that agree on overlaps up to a cocycle glue to a global section iff the Čech class vanishes. This uses $H^1$ with coefficients in the sheaf of gauges, nonabelian $H^1$ for torsors. Extension over a CW complex cell by cell meets obstructions in $H^{k+1}(B; \pi_k(F))$.*

RG's obstruction tower (L131) and the global gluing layer (atlas A6, $H^1 = \mathbb{Z}_2$ for the Möbius family) are instances of these standard results. The print's battery checks numbers. The statements are textbook obstruction theory.

**Imported Theorem 3.4** (Thom's first isotopy lemma [10])**.** *A proper stratified submersion of a Whitney stratified space is locally topologically trivial over each stratum.*

**Corollary 3.5** (Invariants change only at walls)**.** *If the law map of a tame experiment is proper and stratified, every protocol invariant that is a topological invariant of the fibre is constant along each stratum of the base. It can change only across walls, i.e. lower-dimensional strata.*

## 4 Normal forms at walls

**Theorem 4.1** (Codimension and forced monodromy)**.** *In the space of real symmetric $n \times n$ matrices, the set with a repeated eigenvalue has codimension 2. Consequently:*

- *a generic one-parameter family has no crossings, only avoided crossings;*
- *a generic two-parameter family has isolated conical points;*
- *along a small loop around a conical point the tracked eigenline has monodromy $-1$.*

*Proof. Codimension.* Near a double eigenvalue, the degenerate block is

$$\lambda I + \begin{pmatrix} a & b \\ b & -a \end{pmatrix},$$

which is degenerate iff $a = b = 0$: two independent conditions. Transversality gives the generic statements.

*Monodromy.* On the loop $(a, b) = (\cos\varphi, \sin\varphi)$, the eigenvector of eigenvalue $+1$ is

$$v(\varphi) = \left(\cos\tfrac{\varphi}{2},\ \sin\tfrac{\varphi}{2}\right), \qquad v(2\pi) = -v(0),$$

which is verified symbolically (W4). The von Neumann–Wigner argument is classical [11], and RG's L122 is this theorem. □

## 5 Forgetting and chaos: cone contraction and transfer operators

**Imported Theorem 5.1** (Birkhoff–Hopf [12, 13])**.** *Let $K$ map the positive cone into itself, and let $\Delta$ be the projective diameter of its image in Hilbert's metric*

$$d_H(x, y) = \log \frac{\max_i(x_i/y_i)}{\min_i(x_i/y_i)}.$$

*Then $d_H(Kx, Ky) \le \tau\, d_H(x, y)$ with*

$$\tau = \tanh(\Delta/4) = \frac{1 - \sqrt{\phi}}{1 + \sqrt{\phi}}, \qquad \phi = \min \frac{K_{ik}K_{jl}}{K_{jk}K_{il}}.$$

**Corollary 5.2** (Uniform forgetting for forced protocols)**.** *Let a declared coarse protocol apply positive maps $K_1, K_2, \ldots$ that may depend on time or on an external forcing, with $\Delta(K_j) \le \Delta_{\max} < \infty$ uniformly. Then any two initial laws satisfy*

$$d_H(K_n \cdots K_1 x,\ K_n \cdots K_1 y) \le \tanh(\Delta_{\max}/4)^n\, d_H(x, y).$$

*Forgetting is exponential and uniform in the forcing. For a binary symmetric channel $\tau = 1 - 2\varepsilon$ exactly.*

*Proof.* Apply Birkhoff–Hopf at each step and multiply. The binary symmetric case has $\phi = (\varepsilon/(1-\varepsilon))^2$, derived symbolically (W1). In W1, a nine-step forced chain of three fixed rational $3 \times 3$ matrices satisfies the product bound at every step. □

*Meaning for L133.* For any *discretized* protocol of forced dynamics in which each step has bounded projective diameter on the declared resolution, uniform forgetting is a theorem, not a simulation. The open case is exactly the one in which no such bound exists, i.e. deterministic dynamics at full resolution, which is lossless by the Rooting Closure.

**Proposition 5.3** (Doeblin minorization)**.** *If $K(\cdot \mid x) \geq \delta\nu$ for all $x$, with $\delta > 0$ and a law $\nu$, then* $\mathrm{TV}(Kp, Kq) \leq (1-\delta)\,\mathrm{TV}(p, q)$.

*Proof.* Write $K = \delta\,\nu\mathbf{1}^\top + (1-\delta)\tilde{K}$ with $\tilde{K}$ stochastic. The first part sends $p - q$ to 0, and $\tilde{K}$ is a TV contraction. □

**Imported Theorem 5.4** (Lasota–Yorke spectral gap [14, 15])**.** *For piecewise $C^2$ maps of an interval with* $\inf |T'| > 1$*, the transfer operator is quasi-compact on $BV$. If the map is mixing, it has a spectral gap, so correlations of $BV$ observables decay exponentially.*

*Consequence.* For piecewise expanding chaos, forgetting at the resolution of $BV$ observables is exponential by a structural theorem. This generalizes the exact Markov-partition result of the Rooting Closure (doubling map) beyond Markov partitions.

## 6 Ledger: from synthetic batteries to structural statements

| Synthetic support in earlier papers | Structural replacement | Status |
|---|---|---|
| Wall-crossing gauges sampled on rays and circles | Theorem 2.7 (curve selection) | STRUCTURAL (tame) |
| Atlas layers, finite registry | Theorem 2.4 (Hardt, cells) | STRUCTURAL (per family) |
| Mesh closure and coverage step | Corollary 3.2 (nerve theorem) | STRUCTURAL |
| Holonomy counts on loops | Theorem 4.1 plus Corollary 3.2 | STRUCTURAL |
| Global gluing classes ($H^1$) | Čech descent, obstruction theory | STRUCTURAL (imported) |
| Moduli and rates at walls | Theorem 2.9, RLCT invariance | STRUCTURAL (polynomially bounded) |
| Forgetting simulations | Corollary 5.2, Doeblin, Lasota–Yorke | STRUCTURAL (bounded diameter, expanding) |
| Uniform enumeration of tame types (L132) | — | NOT REACHED |
| Deterministic turbulence mixing (L133) | — | NOT REACHED |
| Non-tame (wild) families | No-Go 2.5 | tameness must be declared |

## 7 What this buys, and what it does not

- **Easier to follow, harder to dispute.** A reader who accepts o-minimality, nerves, Čech descent, transversality and Birkhoff–Hopf, all standard, must accept RG's global layer for tame experiments. No synthetic sampling is needed for the claims.

- **One declaration does the work.** Tameness is the single hypothesis behind finiteness, crossing and power laws. It is necessary (No-Go 2.5), and it is satisfied by every semialgebraic, analytic or exponential-family model in use.
- **Not novel.** The frameworks are classical. The contribution is the placement: the RG corollaries and their proofs that RG objects meet the hypotheses.
- **Not reached.** The uniform tame enumeration (L132) and deterministic mixing (L133) remain external.

## 8 Witnesses and limits

- `rg_structural_witnesses.py` runs W1–W6, all exact or symbolic, with no random sampling. Reruns are byte-identical.
- *W1:* the binary symmetric $\tau = 1 - 2\varepsilon$, symbolic, and a nine-step forced chain of exact rationals.
- *W2:* exact nerve cohomology.
- *W3:* limits along rays and along a curve, symbolic.
- *W4:* conical monodromy, symbolic.
- *W5:* power-law moduli, symbolic.
- *W6:* the three types of $t^2 + at + b$, and the accumulating walls of $\sin(1/\theta)$.
- *Limits.* All framework theorems are imported. Tameness, properness and goodness of covers are hypotheses to be declared. The witnesses check constants; the proofs carry the claims.

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